LVDS Display Interface Hardware Design

Name: Namrata Dolas
Role: Hardware Design Intern
Project Title: LVDS Display Connector Interface Board

Tool Used: KiCad 7.0
Board Type: High-speed mixed-signal PCB
Domain: Embedded Hardware

Senior / Reporting Manager: Balaji Mayavan

Date: 12 Jan 2026

1.Project Overview

The objective of this project was to design and validate an LVDS (Low Voltage Differential Signal) display interface PCB, ensuring correct signal integrity, power distribution, EMI robustness, and manufacturing readiness.

The board acts as an interconnection interface between:

- LVDS source connector (J1)
- LVDS display connector (J2)
- Power input (+12V via DC Jack)
- Control signals (Enable, Dimming, I2C, Reset)

The design was carried out from

schematic → ERC→ PCB layout → DRC → documentation

with industry-standard practices.

2.Scope of Work

My responsibilities included:

- Understanding LVDS signal requirements
- Studying connector pinouts and datasheets
- Creating schematic with correct net assignments
- Designing PCB with controlled routing practices
- Solving DRC/ERC issues independently
- Preparing design for fabrication
- Generating BOM and documentation

3.Learning Objectives (Self-Exploration)

During this project, I learned:

- Fundamentals of LVDS differential signal
- Importance of impedance control and length matching
- Correct handling of mounting pads (MP pads)
- Copper clearance vs solder mask rules
- Net connectivity and DRC interpretation
- Real-world PCB debugging workflow
- Manufacturing-oriented footprint configuration

4.LVDS Technical Background

4.1 What is LVDS?

LVDS is a high-speed differential signal standard characterized by:

- Low voltage swing (~350 mV)
- Differential pairs (+/–)
- High noise immunity
- Low EMI emission

4.2 Why Differential Signal?

- Cancels common-mode noise
- Enables high data rates
- Improves signal integrity over longer traces

System Diagram

Figure 1 shows a typical connection with LVDS transmitters and receivers.

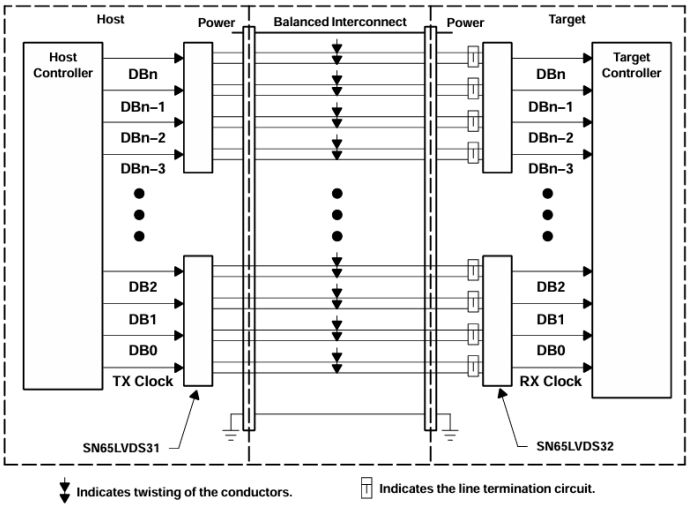
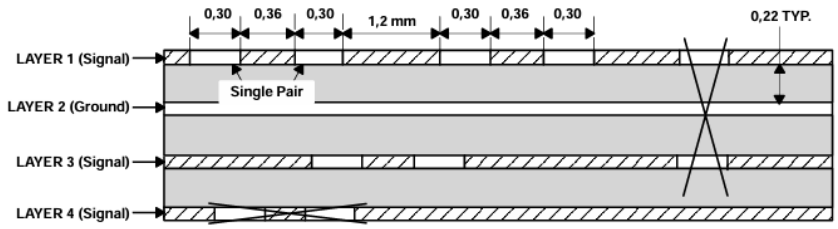


Figure 1. Typical Connection with LVDS Transmitters and Receivers

Interface Definition

The data inputs to the SN65LVDS31 are received at the interface of the PCB traces from the host controller and consist of up to n bits of information and a transmit (Tx) clock.



NOTES:
A. All fabrication item must meet or exceed best industry practice.
B. Laminate material: copper clad FR-4
C. Copper weight: 1 oz. start
D. Finished board thickness: 0.032 (±0.010) inches
E. Dielectric thickness to be symmetrical between all layers (±0.005 inches)
F. Maximum warp and twist: 0.001 inches per inch
G. Circuitry on outer layers to be tin-lead plated (60/40), plated to 300 µin (minimum)
H. Soldermask both sides per artwork: green enthone
I. Copper plating to be 0.001 inches (minimum) in plated-through holes
J. Soldermask over bare copper with tin-lead hot air leveling
K. Dimensions shown in Figure 2 are in millimeters

Figure 2. Typical PCB Construction

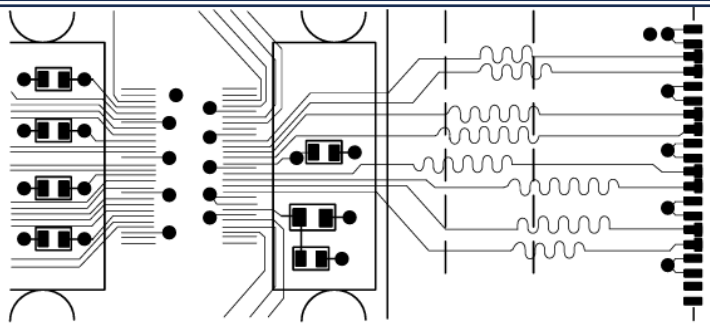


Figure 3. Mitering Output Traces to Closely Match Lengths

LVDS Design ref. - "Low-Voltage Differential Signalling LVDS Design Notes"

5. Schematic Design Details

5.1 Major Components

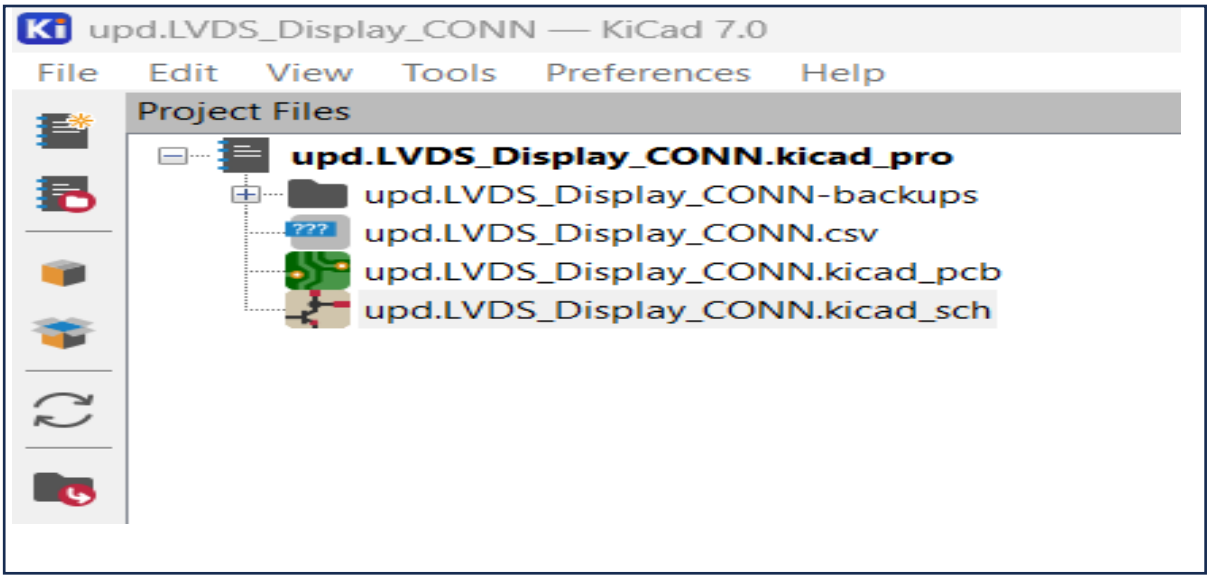
- **J1:** AM62P-LP microprocessor connector

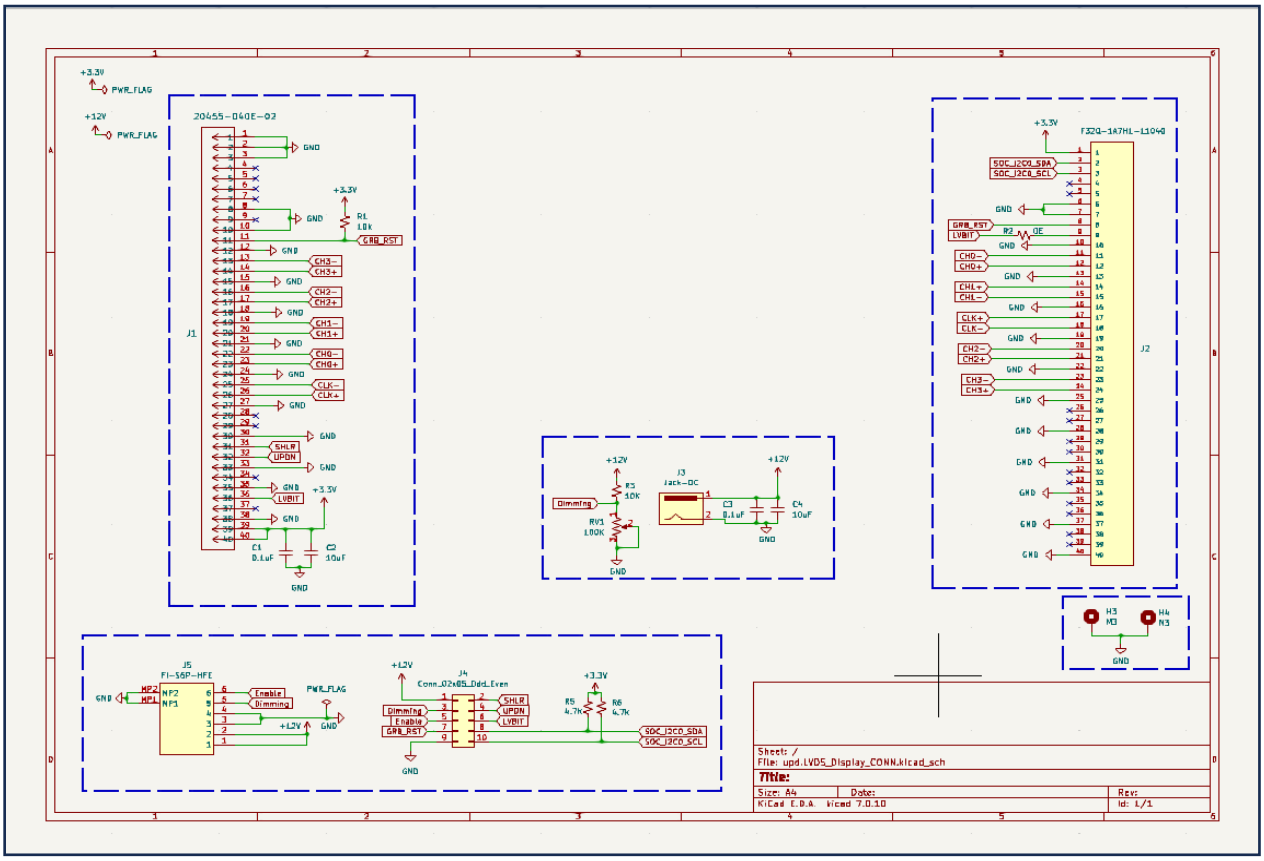
  {LVDS input connector (IPEX 20455-040E-02)}
- **J2:** display connector

  {LVDS output (F32Q-147H-11040)}
- **J3:** DC Barrel Jack (+12V)
- **J5:** Control header (Enable, Dimming, Reset)
- **R/C networks:** Pull-ups, filtering, decoupling

5.2 Power Architecture

- +12V input → DC Jack
- +3.3V logic rail
- Proper decoupling capacitors placed near loads
- Power flags added to satisfy ERC




+3.3V
PWR_FLAG
+12V
PWR_FLAG
20455-040E-02
J1
GND
+3.3V
R1
10k
GRB_RST
CH3-
CH3+
CH2-
CH2+
CH1-
CH1+
CH0-
CH0+
CLK-
CLK+
SHLR
UPDN
LVBIT
C1
0.1uF
C2
10uF
F32Q-1A7H1-11040
SOC_I2C0_SDA
SOC_I2C0_SCL
R2
OE
J2
+1.2V
R3
10K
Dimming
RV1
100K
J3
Jack-DC
C3
0.1uF
C4
10uF
H3
M3
H4
N3
J5
FI-S6P-HFE
MP2
MP1
NP2
NP1
Enable
Dimming
PWR_FLAG
J4
Conn_02x05_Odd_Even
R5
4.7K
R6
4.7K
SOC_I2C0_SDA
SOC_I2C0_SCL
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Title:
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Date:
Rev:
KiCad E.D.A. kicad 7.0.10
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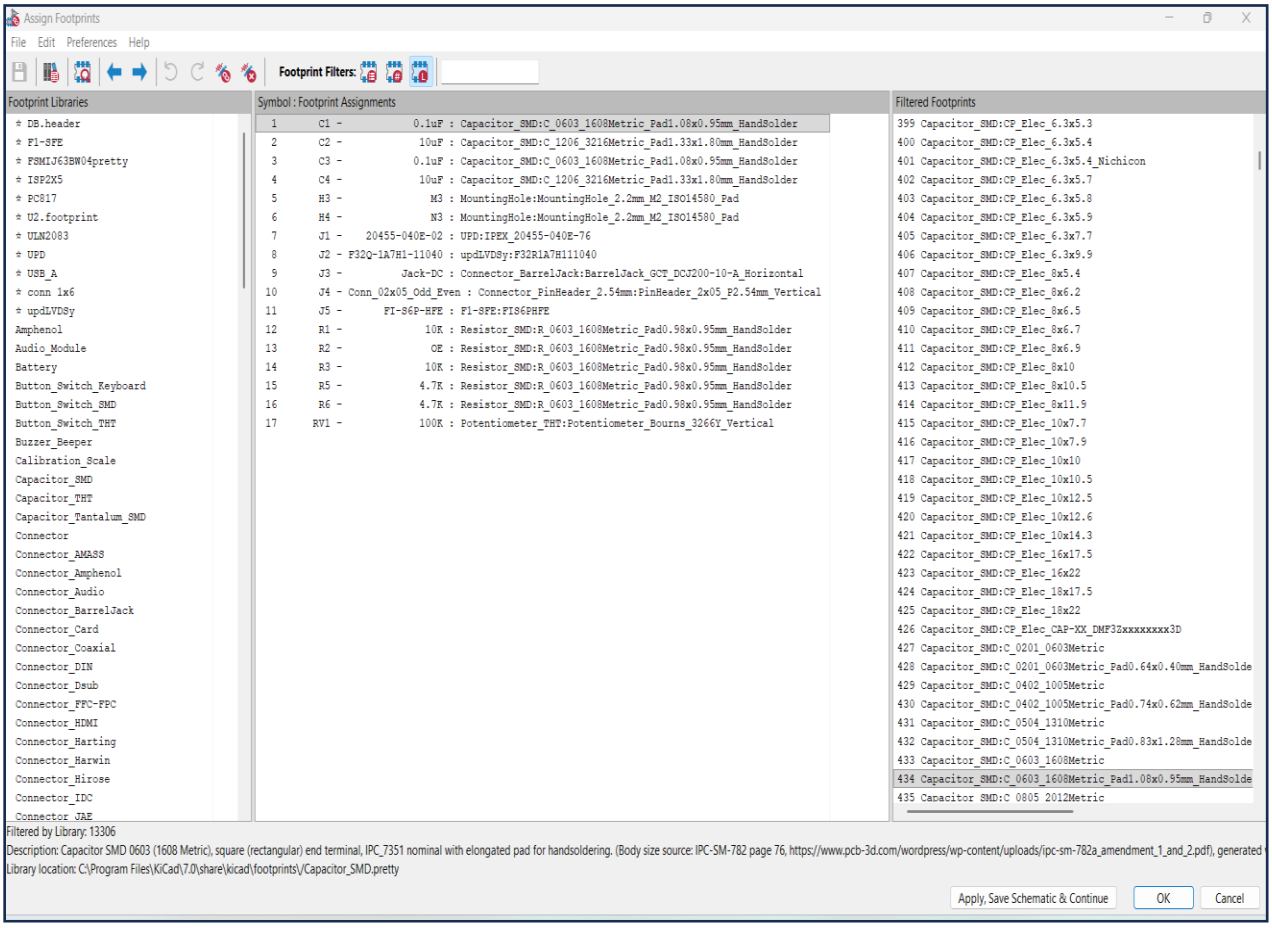
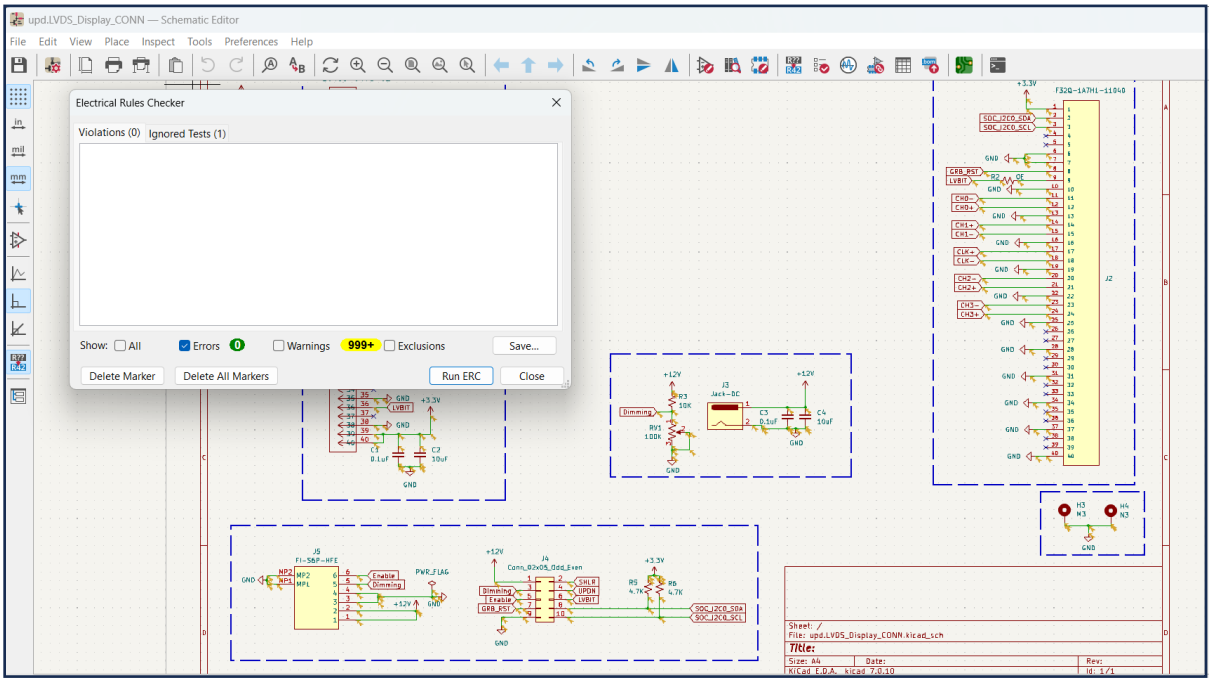

Assign Footprints
File Edit Preferences Help
Footprint Filters:
Footprint Libraries
DB.header
FI-SFE
FSMIJ63BW04pretty
ISP2X5
PC817
U2.footprint
ULN2083
UPD
USB_A
conn 1x6
updLVDSy
Amphenol
Audio_Module
Battery
Button_Switch_Keyboard
Button_Switch_SMD
Button_Switch_THT
Buzzer_Beeper
Calibration_Scale
Capacitor_SMD
Capacitor_THT
Capacitor_Tantalum_SMD
Connector
Connector_AMASS
Connector_Amphenol
Connector_Audio
Connector_BarrelJack
Connector_Card
Connector_Coaxial
Connector_DIN
Connector_Dsub
Connector_FFC-FPC
Connector_HDMI
Connector_Harting
Connector_Harwin
Connector_Hirose
Connector_IDC
Connector_JAE
Symbol : Footprint Assignments
1 C1 - 0.1uF : Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSolder
2 C2 - 10uF : Capacitor_SMD:C_1206_3216Metric_Pad1.33x1.80mm_HandSolder
3 C3 - 0.1uF : Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSolder
4 C4 - 10uF : Capacitor_SMD:C_1206_3216Metric_Pad1.33x1.80mm_HandSolder
5 H3 - M3 : MountingHole:MountingHole_2.2mm_M2_ISO14580_Pad
6 H4 - N3 : MountingHole:MountingHole_2.2mm_M2_ISO14580_Pad
7 J1 - 20455-040E-02 : UPD:IPEX_20455-040E-76
8 J2 - F32Q-1A7H1-11040 : updLVDSy:F32R1A7H111040
9 J3 - Jack-DC : Connector_BarrelJack:BarrelJack_GCT_DCJ200-10-A_Horizontal
10 J4 - Conn_02x05_Odd_Even : Connector_PinHeader_2.54mm:PinHeader_2x05_P2.54mm_Vertical
11 J5 - FI-S6P-HFE : FI-SFE:FIS6PHFE
12 R1 - 10K : Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSolder
13 R2 - OE : Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSolder
14 R3 - 10K : Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSolder
15 R5 - 4.7K : Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSolder
16 R6 - 4.7K : Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSolder
17 RV1 - 100K : Potentiometer_THT:Potentiometer_Bourns_3266Y_Vertical
Filtered Footprints
399 Capacitor_SMD:CP_Elec_6.3x5.3
400 Capacitor_SMD:CP_Elec_6.3x5.4
401 Capacitor_SMD:CP_Elec_6.3x5.4_Nichicon
402 Capacitor_SMD:CP_Elec_6.3x5.7
403 Capacitor_SMD:CP_Elec_6.3x5.8
404 Capacitor_SMD:CP_Elec_6.3x5.9
405 Capacitor_SMD:CP_Elec_6.3x7.7
406 Capacitor_SMD:CP_Elec_6.3x9.9
407 Capacitor_SMD:CP_Elec_8x5.4
408 Capacitor_SMD:CP_Elec_8x6.2
409 Capacitor_SMD:CP_Elec_8x6.5
410 Capacitor_SMD:CP_Elec_8x6.7
411 Capacitor_SMD:CP_Elec_8x6.9
412 Capacitor_SMD:CP_Elec_8x10
413 Capacitor_SMD:CP_Elec_8x10.5
414 Capacitor_SMD:CP_Elec_8x11.9
415 Capacitor_SMD:CP_Elec_10x7.7
416 Capacitor_SMD:CP_Elec_10x7.9
417 Capacitor_SMD:CP_Elec_10x10
418 Capacitor_SMD:CP_Elec_10x10.5
419 Capacitor_SMD:CP_Elec_10x12.5
420 Capacitor_SMD:CP_Elec_10x12.6
421 Capacitor_SMD:CP_Elec_10x14.3
422 Capacitor_SMD:CP_Elec_16x17.5
423 Capacitor_SMD:CP_Elec_16x22
424 Capacitor_SMD:CP_Elec_18x17.5
425 Capacitor_SMD:CP_Elec_18x22
426 Capacitor_SMD:CP_Elec_CAP-XX_DMP32xxxxxxxx3D
427 Capacitor_SMD:C_0201_0603Metric
428 Capacitor_SMD:C_0201_0603Metric_Pad0.64x0.40mm_HandSolde
429 Capacitor_SMD:C_0402_1005Metric
430 Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSolde
431 Capacitor_SMD:C_0504_1310Metric
432 Capacitor_SMD:C_0504_1310Metric_Pad0.83x1.28mm_HandSolde
433 Capacitor_SMD:C_0603_1608Metric
434 Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSolde
435 Capacitor_SMD:C_0805_2012Metric
Filtered by Library: 13306
Description: Capacitor SMD 0603 (1608 Metric), square (rectangular) end terminal, IPC_7351 nominal with elongated pad for handsoldering. (Body size source: IPC-SM-782 page 76, https://www.pcb-3d.com/wordpress/wp-content/uploads/ipc-sm-782a_amendment_1_and_2.pdf), generated
Library location: C:\Program Files\KiCad\7.0\share\kicad\footprints\/Capacitor_SMD.pretty
Apply, Save Schematic & Continue
OK
Cancel

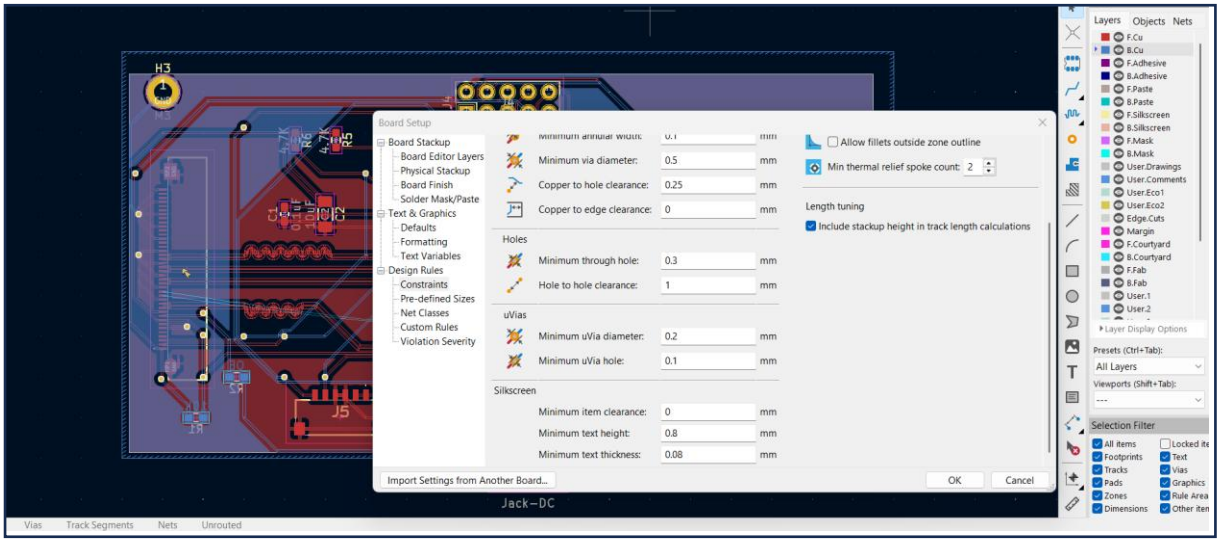
Source:	C:\Users\MANSI\Downloads\upd.LVDS_Display_CONN\upd.LVDS_Display_CONN.kicad_sch				
Date:	12-01-2026 09:43				
Tool:	Eeschema 7.0.10				
Generator:	C:\Program Files\KiCad\7.0\bin\scripting\plugins\bom_csv_grouped_by_value.py				
Component	17				
Collated Components:					
Item	Qty	Reference(s	Value	LibPart	Footprint
1	2	C1, C3	0.1uF	Device:C_Small	Capacitor_SMD:C_0603_1608Metric_Pad1.08x0
2	2	C2, C4	10uF	Device:C_Small	Capacitor_SMD:C_1206_3216Metric_Pad1.33x1
3	1	H3	M3	Mechanical:MountingHole_Pad	MountingHole:MountingHole_2.2mm_M2_ISO1
4	1	H4	N3	Mechanical:MountingHole_Pad	MountingHole:MountingHole_2.2mm_M2_ISO1
5	1	J1	20455-040E-02	updLVDS:20455-040E-02	UPD:IPEX_20455-040E-76
6	1	J2	F32Q-1A7H1-11040	updLVDS:F32Q-1A7H1-11040	updLVDSy:F32R1A7H111040
7	1	J3	Jack-DC	Connector:Jack-DC	Connector_BarrelJack:BarrelJack_GCT_DCJ200-
8	1	J4	Conn_02x05_Odd_Even	Connector_Generic:Conn_02x05_Odd_Even	Connector_PinHeader_2.54mm:PinHeader_2x0
9	1	J5	FI-S6P-HFE	updLVDS:FI-S6P-HFE	F1-SFE:FIS6PHFE
10	2	R1, R3	10K	Device:R_Small_US	Resistor_SMD:R_0603_1608Metric_Pad0.98x0.
11	1	R2	OE	Device:R_Small_US	Resistor_SMD:R_0603_1608Metric_Pad0.98x0.
12	2	R5, R6	4.7K	Device:R_Small_US	Resistor_SMD:R_0603_1608Metric_Pad0.98x0.
13	1	RV1	100K	Device:R_Potentiometer_US	Potentiometer_THT:Potentiometer_Bourns_32

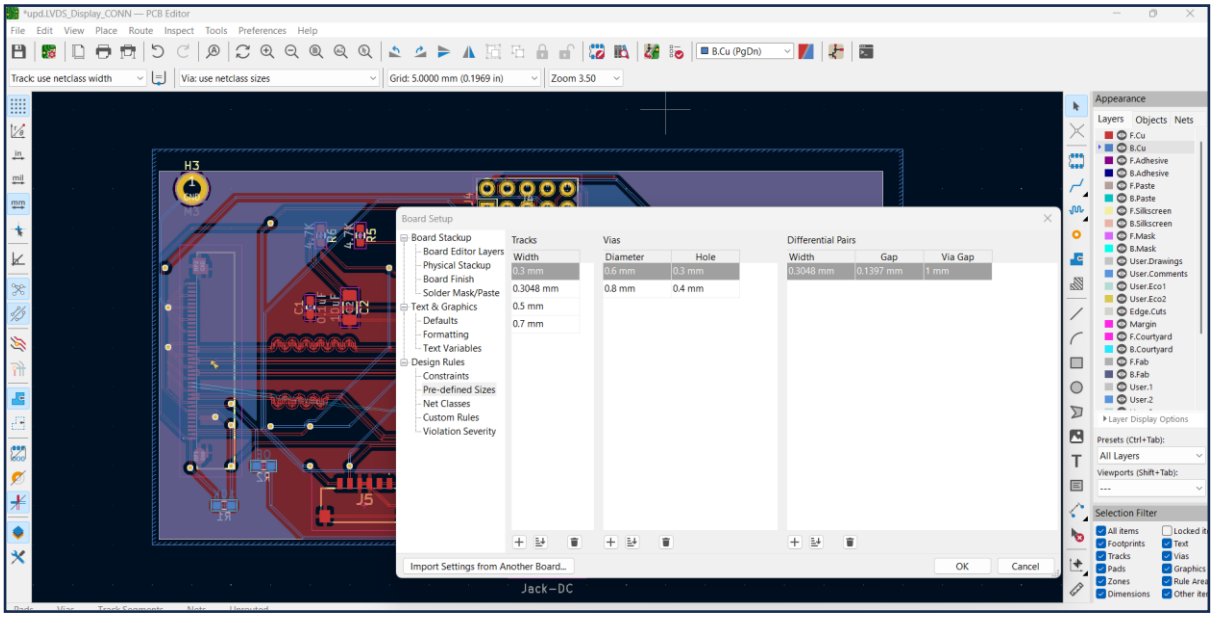


6. PCB Design Workflow

6.1 Board Setup

- Defined net classes
- Set clearance rules (0.15 mm)
- Separate signal and power routing strategy



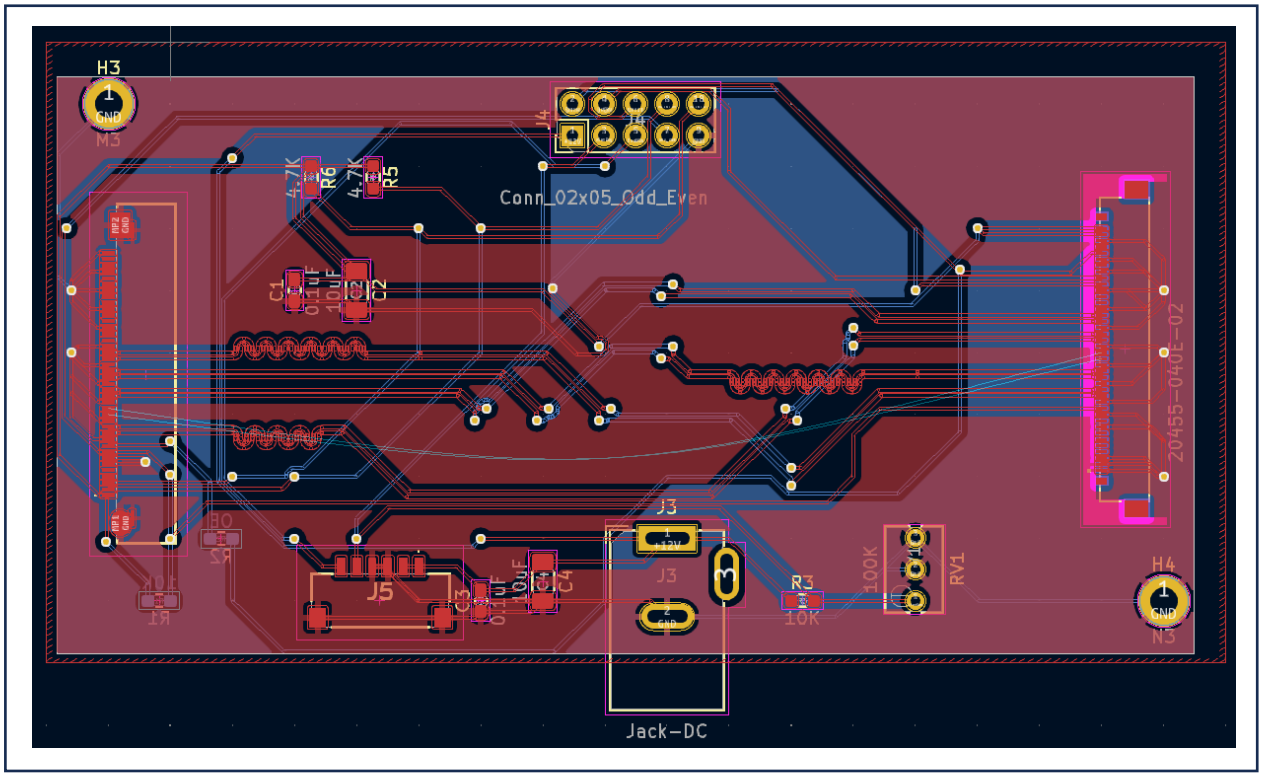


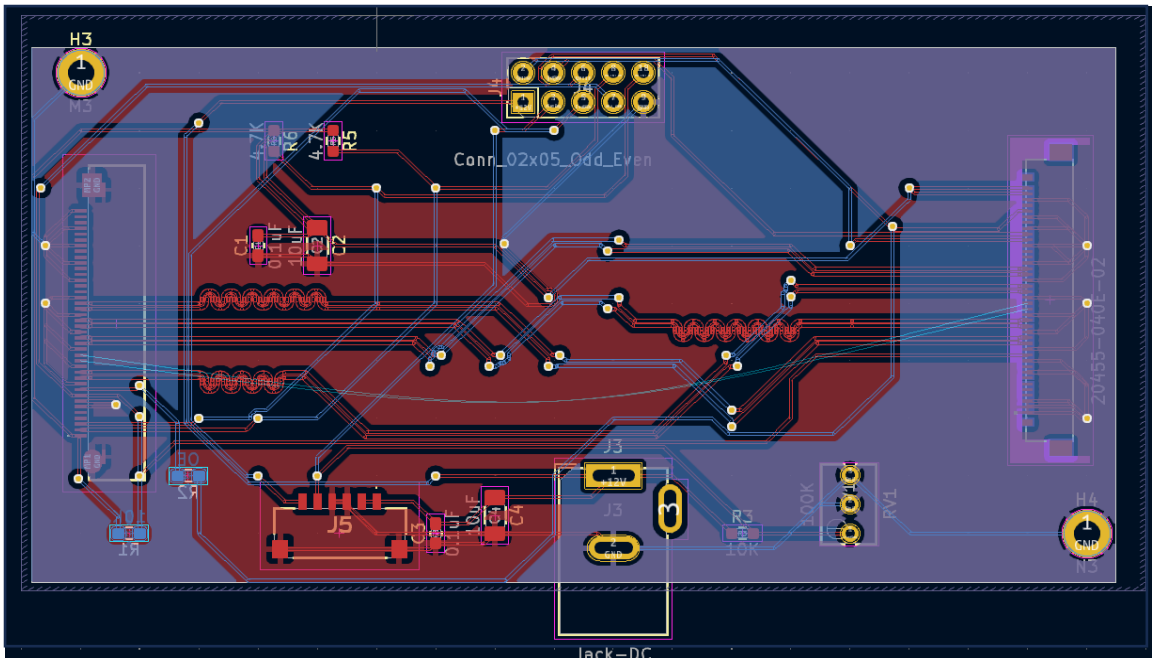
6.2 LVDS Routing Strategy

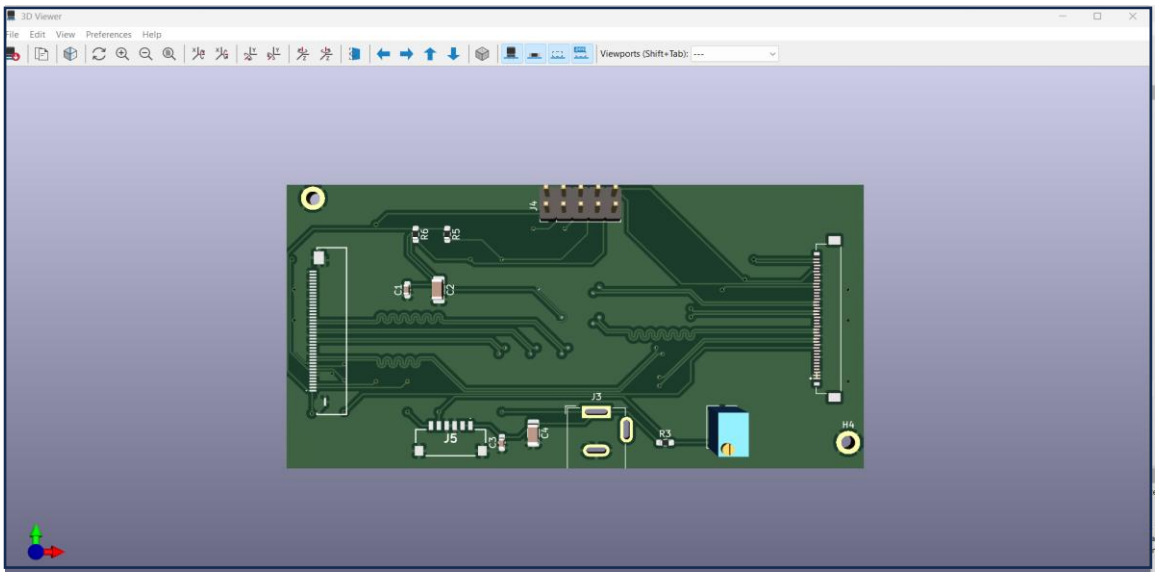
- Differential pairs routed together
- Matched lengths using meanders
- Avoided sharp 90° bends
- Maintained continuous reference plane (GND)

6.3 Grounding Strategy

- Full GND plane on inner/outer layer (Front and back copper layer-→ B.cu & F.cu)
- Stitching vias used near signal transitions
- Connector mounting pads tied to GND (MP1 , MP2 , MP3 , MP4)




H3
GND
Conn_02x05_Odd_Even
J4
R6
4.7K
R5
4.7K
C1
0.1uF
C2
10uF
J5
C3
0.1uF
C4
10uF
J3
R3
10K
100K
RV1
H4
GND
Jack-DC
20455-040E-02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File Edit View Preferences Help
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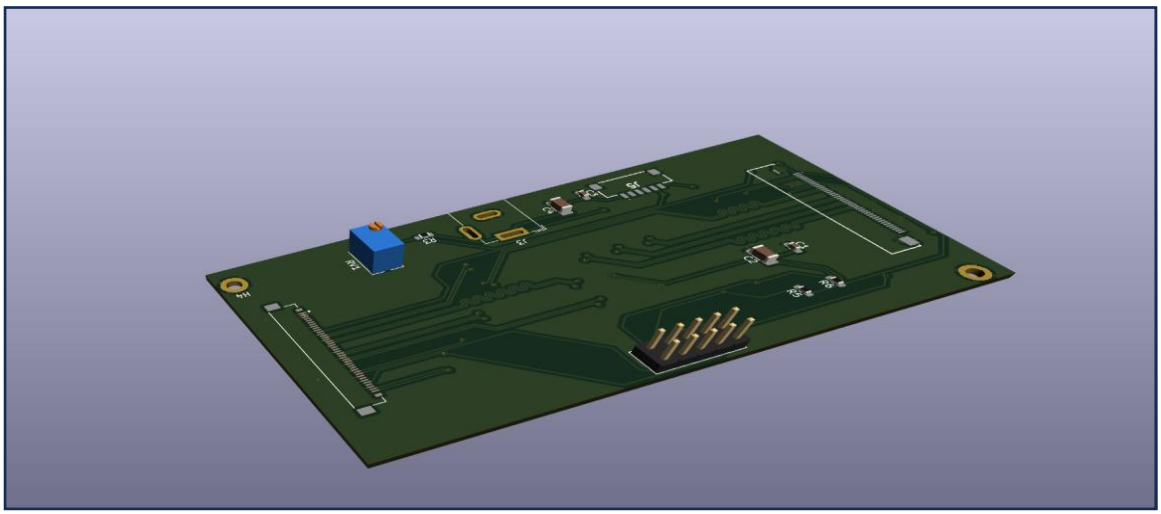


7. Key Design Decisions & Justifications

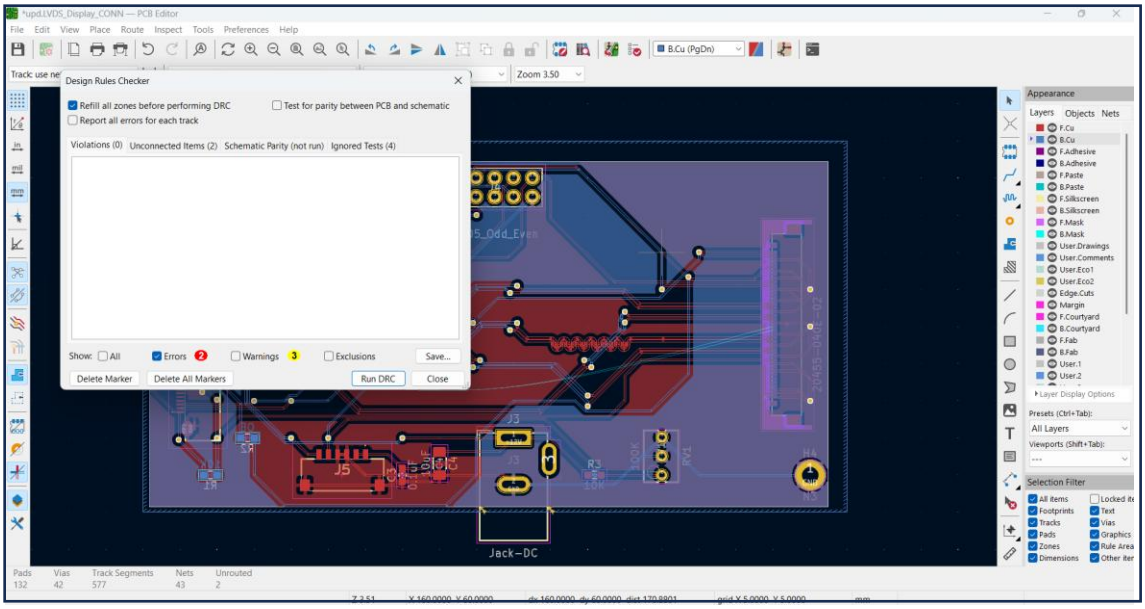
7.1 Mounting Pads (MP Pads)

Decision: MP pads connected to GND
Reason:

- Eliminates copper clearance violations
- Improves EMI shielding
- Provides mechanical stability
- Standard industry practice

7.2 Solder Mask & Paste Control

- MP pads: No solder paste, no solder mask opening
- Signal pads: Standard mask openings
- Prevents solder bridging and floating connectors



8. Challenges Faced & Solutions

8.1 Solder Mask Aperture Bridge Errors

Issue:
DRC reported solder mask bridging between MP pads.

Root Cause:
Overlapping solder mask openings on adjacent MP pads.

Solution:
Disabled F. Mask and F.Paste on MP pads.

8.2 Clearance Violations (0.000 mm)

Issue:
Copper zones touching MP pads.

Root Cause:
MP pads were <no net> while GND pour surrounded them.

Solution:
Assigned MP pads to GND net and refilled zones.

8.3 <no net> Tracks Causing DRC Errors

Issue:
Tracks visually connected but electrically unassigned.

Solution:
Manually assigned correct nets or removed redundant copper.

9. Verification & Validation

9.1 ERC (Schematic)

- ERC errors: **0**
- Power flags correctly applied

9.2 DRC (PCB)

- Clearance violations: **0**
- Unconnected items: **0**
- Solder mask errors: **0**

9.3 Manufacturability Check

- Courtyard clearances respected
- Proper drill sizes
- Mask and paste layers verified
- Fabrication-ready

10. Bill of Materials (BOM)

- BOM generated using **KiCad grouped by value**
- Included:
  - Reference
  - Quantity
  - Value
  - Footprint
  - Library source

	A	B	C	D	E	F
1	Source:	C:\Users\MANSI\Downloads\upd.LVDS_Display_CONN\upd.LVDS_Display_CONN.kicad_sch				
2	Date:	12-01-2026 09:43				
3	Tool:	Eeschema 7.0.10				
4	Generator:	C:\Program Files\KiCad\7.0\bin\scripting\plugins/bom_csv_grouped_by_value.py				
5	Component	17				
6						
7	Collated Components:					
8						
9	Item	Qty	Reference(s	Value	LibPart	Footprint
10	1	2	C1, C3	0.1uF	Device:C_Small	Capacitor_SMD:C_0603_1608Metric_Pad1.08x(
11	2	2	C2, C4	10uF	Device:C_Small	Capacitor_SMD:C_1206_3216Metric_Pad1.33x1
12	3	1	H3	M3	Mechanical:MountingHole_Pad	MountingHole:MountingHole_2.2mm_M2_ISO1
13	4	1	H4	N3	Mechanical:MountingHole_Pad	MountingHole:MountingHole_2.2mm_M2_ISO1
14	5	1	J1	20455-040E-02	updLVDS:20455-040E-02	UPD:IPEX_20455-040E-76
15	6	1	J2	F32Q-1A7H1-11040	updLVDS:F32Q-1A7H1-11040	updLVDSy:F32R1A7H111040
16	7	1	J3	Jack-DC	Connector:Jack-DC	Connector_BarrelJack:BarrelJack_GCT_DCJ200-
17	8	1	J4	Conn_02x05_Odd_Even	Connector_Generic:Conn_02x05_Odd_Even	Connector_PinHeader_2.54mm:PinHeader_2x0
18	9	1	J5	FI-S6P-HFE	updLVDS:FI-S6P-HFE	F1-SFE:FIS6PHFE
19	10	2	R1, R3	10K	Device:R_Small_US	Resistor_SMD:R_0603_1608Metric_Pad0.98x0.
20	11	1	R2	OE	Device:R_Small_US	Resistor_SMD:R_0603_1608Metric_Pad0.98x0.
21	12	2	R5, R6	4.7K	Device:R_Small_US	Resistor_SMD:R_0603_1608Metric_Pad0.98x0.
22	13	1	RV1	100K	Device:R_Potentiometer_US	Potentiometer_THT:Potentiometer_Bourns_32

upd.LVDS_Display_CONN

11. Tools & Resources Used

11.1 Software

- KiCad 7.0
- KiCad BOM Generator

11.2 Documentation & Learning Resources

- Connector datasheets
- LVDS application notes
- IPC routing guidelines
- Self-debugging via DRC interpretation
- Mentor-guided design review feedback

12. Key Takeaways (Professional Learning)

- Visual connection ≠ electrical connection
- DRC is a design validation tool, not an obstacle
- High-speed design demands discipline and consistency
- Small errors (0.025 mm) matter in fabrication
- Real hardware design involves iteration, not perfection in first attempt

13. Final Project Status

- Schematic complete
- PCB routed
- DRC/ERC clean
- BOM generated
- Ready for fabrication

14. Conclusion

This project strengthened my understanding of real-world hardware design, especially high-speed interfaces like LVDS. Through hands-on debugging, self-learning, and systematic validation, I successfully completed a manufacturable LVDS interface PCB aligned with industry standards.